

Multi-Stream Lehmer Random Number Generators

Multi-stream Lehmer random number generators (RNGs) are designed to produce multiple independent sequences of pseudo-random numbers, essential for parallel computing, reproducibility, and synchronized simulations. These generators extend the classic Lehmer RNG (also known as a multiplicative linear congruential generator) by partitioning its period into distinct streams and substreams.

Multi-Stream Implementations

To support multiple streams:

- **Stream Initialization:** Each stream starts at a seed value spaced sufficiently apart within the generator's period. For example, Steve Park's ANSI C library uses 256 streams, selectable via `SelectStream(s)`.
- **Substream Partitioning:** Streams can be further divided into substreams.
- **State Management:** Libraries like `RngStreams` allow resetting streams to their initial state, advancing to the next substream, or rewinding.

Applications and Considerations

- **Parallel Simulations:** Independent streams enable parallelized Monte Carlo experiments without overlap.
- **Reproducibility:** Seeds and stream indices ensure repeatable results.
- **Statistical Quality:** Properly spaced streams avoid correlation, critical for validity.